

ELEC2134: Class Test 1 (Fourier Series)

Date: Thursday 26 February 2026 (Week 2)

Time: 3 - 4 pm (Exam Duration 40 minutes).

Instruction to candidates: To maximise your marks, show all steps clearly when deriving your answers (do not write final answers only).

Pass Level Questions. Total Marks: 64

Q1: For the periodic voltage waveform shown in Figure 1,

- Show that the expression for $v(t)$ over $0 \leq t \leq T$ is given by
[4 Marks]
$$v(t) = \frac{V_m}{T} t$$
- Show that the Fourier Coefficient $A_0 = \frac{V_m}{2}$
[5 Marks]
- Show that the Fourier Coefficient $B_n = \frac{-V_m}{\pi n}$
[20 Marks]
- Write the Fourier series for $v(t)$ up to and including the third harmonic.
[15 Marks]

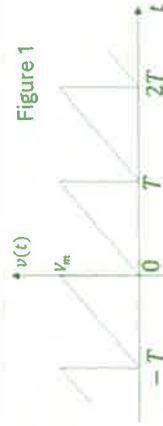


Figure 1

Q2: The periodic function $f(t)$ is known to have even symmetry and the amplitude spectrum is shown in Figure 2. If all coefficients A_n and B_n are nonnegative, determine the Fourier series $f(t)$. [20 Marks]



Figure 2

Higher Level Questions. Total Marks: 36

Q3: Figure 3 shows one period of a periodic waveform with period $T = 2$ s.

- Sketch the complete waveform over two periods, i.e. for $-2 \leq t \leq 2$ s, assuming the signal satisfies even symmetry and half-wave symmetry. [12 Marks]
- Determine the complex Fourier series coefficients C_n for your sketched waveform in part (a) above, and show that
[24 Marks]

$$C_n = \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \quad \text{for } n \text{ odd.}$$

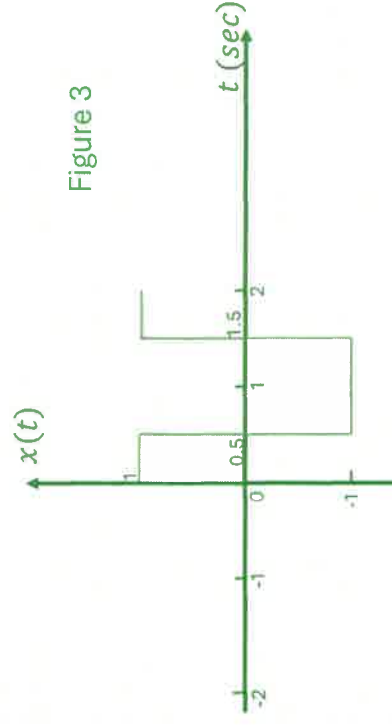


Figure 3

1. Integrals (for integer $n \neq 0$)

$$\int \sin(ax) dx = -\frac{1}{a} \cos(ax); \quad \int \cos(ax) dx = \frac{1}{a} \sin(ax)$$

$$\int_0^T \sin(n\omega_0 t) dt = 0; \quad \int_0^T \cos(n\omega_0 t) dt = 0$$

2. Integration by Parts (Key Formula)

$$\int u dv = uv - \int v du$$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1); \quad \int x^2 e^{ax} dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

3. Integration by Parts: (a) $\int t \cos(n\omega_0 t) dt$

$$\text{Let } u = t \Rightarrow du = dt; dv = \cos(n\omega_0 t) dt \Rightarrow v = \frac{1}{n\omega_0} \sin(n\omega_0 t)$$

$$\int t \cos(n\omega_0 t) dt = \frac{t}{n\omega_0} \sin(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \cos(n\omega_0 t) + \text{Constant}$$

4. Integration by Parts: (b) $\int t \sin(n\omega_0 t) dt$

$$\text{Let } u = t \Rightarrow du = dt; dv = \sin(n\omega_0 t) dt \Rightarrow v = -\frac{1}{n\omega_0} \cos(n\omega_0 t)$$

$$\int t \sin(n\omega_0 t) dt = -\frac{t}{n\omega_0} \cos(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \sin(n\omega_0 t) + \text{Constant}$$

5. Definite Integration by Parts: For $0 \leq t \leq T$ and integer n :

$$\int_0^T t \sin(n\omega_0 t) dt = \left[\frac{T}{n\omega_0} \right] \quad \text{and} \quad \int_0^T t \cos(n\omega_0 t) dt = \left[0 \right]$$

Useful formulae

6. Definite Integrals

$$\int_0^T x^2 \cos(nx) dx = \left[\frac{x^2}{n} \sin(nx) + \frac{2x}{n^2} \cos(nx) - \frac{2}{n^3} \sin(nx) \right]_0^T$$

$$\int_0^T x^2 \sin(nx) dx = \left[-\frac{x^2}{n} \cos(nx) + \frac{2x}{n^2} \sin(nx) - \frac{2}{n^3} \cos(nx) \right]_0^T$$

7. Definite Integrals

$$\int_0^T x \cos(nx) dx = \left[\frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^T$$

$$\int_0^T x \sin(nx) dx = \left[-\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \right]_0^T$$

8. Definite Integrals

$$\int_0^T \cos(m\omega_0 t) \sin(n\omega_0 t) dt = 0 \quad \text{for all } m \text{ and } n$$

$$\int_0^T \sin(m\omega_0 t) \sin(n\omega_0 t) dt = \begin{cases} \frac{T}{2} & \text{for all } m \neq n \\ 0 & \text{for } m = n \end{cases}$$

$$\int_0^T \cos(m\omega_0 t) \cos(n\omega_0 t) dt = \begin{cases} \frac{T}{2} & \text{for all } m \neq n \\ 0 & \text{for } m = n \end{cases}$$

Useful formulae for Fourier series

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t);$$

$$A_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad A_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$B_n = \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt$$

$$\text{Odd function: } x(t) = -x(-t)$$

$$A_n = 0 \text{ for all } n$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

$$\text{Half-wave symmetry: } x(t) = -x\left(t - \frac{T}{2}\right)$$

$$A_n = 0 \text{ for even } n; \quad B_n = 0 \text{ for even } n$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \text{ for odd } n$$

$$B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \text{ for odd } n$$

$$\text{Even function: } x(t) = x(-t)$$

$$B_n = 0 \text{ for all } n$$

$$A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n)$$

$$K_n \angle \phi_n = A_n - jB_n;$$

$$K_n = \sqrt{A_n^2 + B_n^2}; \quad \phi_n = -\tan^{-1} \left(\frac{B_n}{A_n} \right)$$

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta)$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin^2 \alpha = \frac{1}{2} (1 - \cos 2\alpha)$$

$$\cos^2 \alpha = \frac{1}{2} (1 + \cos 2\alpha)$$

$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left(\alpha + \tan^{-1} \left(\frac{B}{A} \right) \right)$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{2} (A_n - jB_n)$$

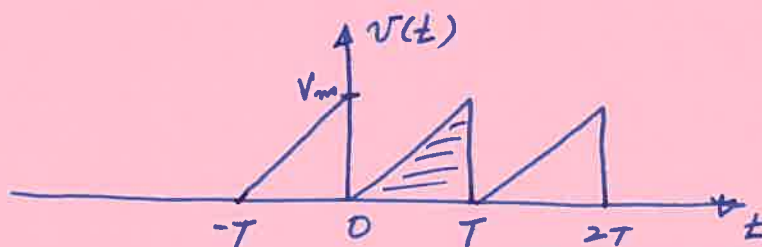
$$\text{For Even Symmetry:}$$

$$C_n = \frac{1}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$$

$$\text{For Odd Symmetry:}$$

$$C_n = \frac{-2j}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

Q1



(a) slope $m = \frac{V_m}{T}$

$\therefore v(t) = \left(\frac{V_m}{T}\right) t$

4 Marks

(b) A_0 (Average Value) = $\frac{\text{Area}}{T} = \frac{\frac{1}{2} T \times V_m}{T} = \frac{V_m}{2}$ } Method 1

5 Marks

OR $A_0 = \frac{1}{T} \int_0^T v(t) dt = \frac{1}{T} \int_0^T \frac{V_m}{T} t dt$ } or

$= \frac{V_m}{T^2} \int_0^T t dt = \frac{V_m}{T^2} \left[\frac{t^2}{2} \right]_0^T$ }

$= \frac{V_m}{T^2} \times \frac{T^2}{2} = \frac{V_m}{2}$

Method 2
5 Marks

(c) $B_n = \frac{2}{T} \int_0^T \frac{V_m}{T} t \sin(n\omega_0 t) dt$

2 Marks

$\omega_0 = \frac{2\pi}{T}$

2 Marks

$= \frac{2V_m}{T^2} \int_0^T t \sin(n\omega_0 t) dt$

$= \frac{2V_m}{T^2} \left\{ \left[\frac{\sin n\omega_0 t}{(n\omega_0)^2} - \frac{t \cos(n\omega_0 t)}{n\omega_0} \right]_0^T \right\}$

6 Marks

$= \frac{2V_m}{T^2} \left\{ \frac{\sin 2\pi/n - 0}{(n\omega_0)^2} - \frac{T \cos(2n\pi) - 0}{n\omega_0} \right\}$

6 Marks

$= \frac{2V_m}{T^2} \times \frac{-T}{n \times \frac{2\pi}{T}} = \frac{2V_m}{T^2} \times \frac{-T^2}{n \times 2\pi}$

4 Marks

$B_n = -\frac{V_m}{\pi n}$

$$V(t) = \frac{V_m}{2} \sum_{n=1}^{\infty} -\frac{V_m}{\pi n} \sin(n\omega_0 t) \quad \leftarrow 5 \text{ Marks} \quad (2)$$

$$V(t) = \frac{V_m}{2} - \frac{V_m}{\pi} \sin(\omega_0 t) - \frac{V_m}{2\pi} \sin(2\omega_0 t) - \frac{V_m}{3\pi} \sin(3\omega_0 t) + \dots$$

10 Marks

Q₂ Even symmetry implies $B_n = 0$ (5 Marks)

$$\therefore \sqrt{A_n^2 + B_n^2} = \sqrt{A_n^2 + 0} = A_n \quad (4 \text{ Marks})$$

From the spectrum $A_0 = 0.8$, $A_1 = 0.2$, $A_2 = 0.6$ & $A_3 = 0.4$

$$f(t) = A_0 + \sum_{n=1}^3 A_n \cos(n\omega_0 t) \quad \omega_0 = 2\pi 500 \quad (5 \text{ Marks})$$

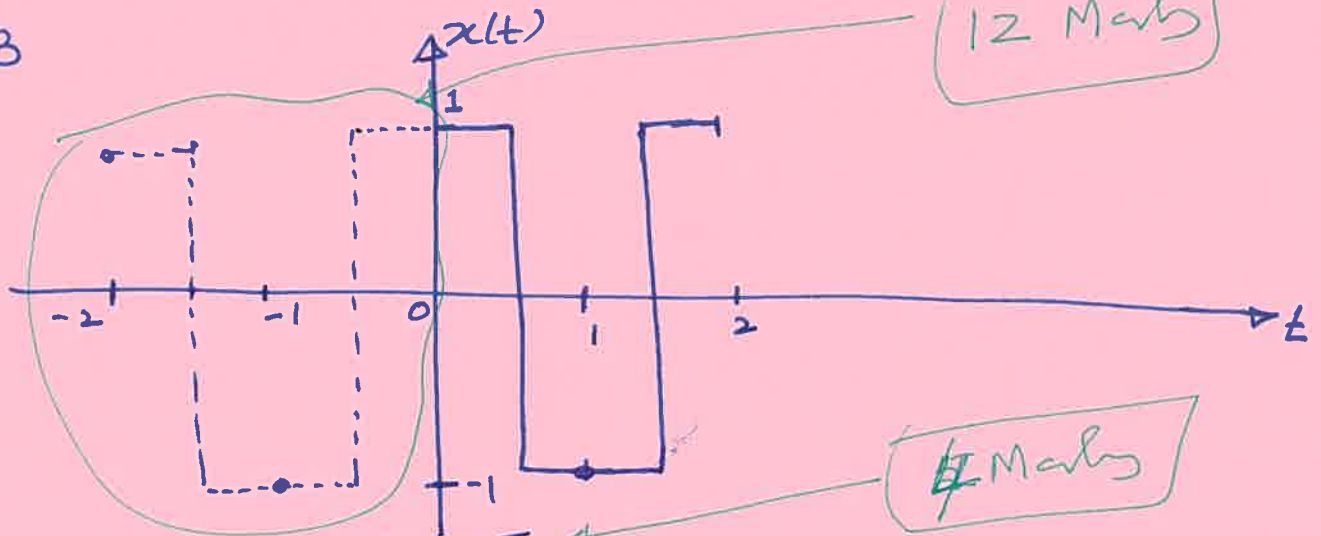
$$\therefore f(t) = 0.8 + 0.2 \cos(1 \times 2\pi \times 500 t) + 0.6 \cos(2 \times 2\pi \times 500 t) + 0.4 \cos(3 \times 2\pi \times 500 t)$$

$$f(t) = 0.8 + 0.2 \cos(1000\pi t) + 0.6 \cos(2000\pi t) + 0.4 \cos(3000\pi t)$$

5 Marks

Higher level questions

Q3



12 Marks

4 Marks

$$T=2 \therefore \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

(b) $C_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jn\pi t} dt$

$$= \frac{1}{2} \left[\int_{-1}^{-1/2} (-1) e^{-jn\pi t} dt + \int_{-1/2}^{1/2} 1 e^{-jn\pi t} dt + \int_{1/2}^1 (-1) e^{-jn\pi t} dt \right]$$

$$= \frac{1}{2} \left\{ (-1) \left[\frac{e^{-jn\pi t}}{-jn\pi} \right]_{-1}^{-1/2} + \left[\frac{e^{-jn\pi t}}{-jn\pi} \right]_{-1/2}^{1/2} - \left[\frac{e^{-jn\pi t}}{-jn\pi} \right]_{1/2}^1 \right\}$$

$$= \frac{1}{2jn\pi} \left(e^{jn\pi/2} \cancel{e^{-jn\pi/2}} + e^{-jn\pi/2} + e^{jn\pi/2} - \cancel{e^{-jn\pi/2}} - e^{-jn\pi/2} \right)$$

$$= \frac{1}{2jn\pi} \left(2e^{jn\pi/2} - 2e^{-jn\pi/2} \right)$$

$$= \frac{2}{n\pi} \left(\frac{e^{jn\pi/2} - e^{-jn\pi/2}}{2j} \right)$$

$$= \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

10 Marks

10 Marks